

WestQuant Representation Stack (Combined)

Slide 1 — Project Overview

Problem: Quantum algorithm design typically performs one fixed translation from optimization problem to hardware. We argue representation itself should be searched.

Approach: A composable pipeline: $P \rightarrow H_i$ (Hamiltonian Explorer) $\rightarrow R_{ij}$ (Representation Scheduler) $\rightarrow Q_{ijk}$ (QoolQit compilation + emulation) $\rightarrow$ Pareto selection. Both projects are independently useful and composable.

Results: Flagship experiment (6 problems $\times$ 5 H $\times$ 9 R $\times$ 3 replicates = 810 cells): H $\times$ R interaction dominates ($\eta^2 = 48.2\%$), embedding main effect $\eta^2(R) = 19.0\%$, Hamiltonian main effect $\eta^2(H) = 6.8\%$. For 5/6 problems, the preselected baseline fails terminal ground-state preservation; search finds feasible alternatives for 6/6. Representation search can rescue an otherwise invalid physical realization.

Importance: The H $\times$ R interaction is the dominant effect, directly motivating joint representation search. Neither H alone nor R alone determines performance — it is their interaction that matters. QoolQit serves as the evaluation engine; WestQuant AI is optional. Fully open-source, deterministic, reproducible.

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Slide 2 — QoolQit Experience

Positive: QoolQit provides a clean end-to-end stack: Register, Drive, QuantumProgram, compile_to(device), LocalEmulator. The embedding API (InteractionEmbedder, SpringLayoutEmbedder, Blade) covers the main approaches. DataGraph bridges logical and physical.

Challenges: DMM requires AnalogDeviceWithDMM (not AnalogDevice). max_energy profile needed for auto-rescaling. Bitstring results are Counters of strings. Native Rydberg $J > 0$ constrains which Hamiltonians are directly realizable. InteractionEmbedder x0=None uses a fixed internal seed.

Workflow: Successive halving (cheap metrics -> emulation) is essential for search scalability. The five-class equivalence taxonomy ensures scientific rigor. Two-way ANOVA with H, R, H×R, and residual terms quantifies how each representation layer matters.